

Candidate's name CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS 2018

Answers

CHEMISTRY
HIGHER 2

9729/4
6 August 2018
2 hours 30 minutes

Paper 4 Practical Test

Candidates answer on the Question Paper.
No Additional Materials are required.

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READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces provided on this cover page.

Give details of the practical shift and laboratory, where appropriate, in the box provided.

Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

Qualitative Analysis Notes are printed on pages 18 and 19.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

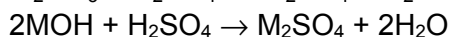
Shift
Laboratory

For Examiner's Use	
1	/41
2	/14
Total	
/ 55	

Answer **all** the questions in the spaces provided.

1 Determination of concentration of sulfuric acid

Both group 1 carbonates and hydroxides are bases which neutralise sulfuric acid as shown by the equations below.



FA 1 is dilute solution of sulfuric acid, H_2SO_4

FA 2 is $0.550 \text{ mol dm}^{-3}$ sodium hydroxide, NaOH

FA 3 is anhydrous powder of an unknown group 1 carbonate

In this question, you will perform an experiment to determine the concentration of the sulfuric acid solution by means of a graphical method and hence, determine the relative atomic mass of an unknown group 1 element.

You will first prepare a standard solution of the group 1 carbonate, **FA 4**, using its anhydrous powder, **FA 3**.

You will then add different volumes of **FA 4** to identical samples of dilute sulfuric acid, **FA 1**. In each case, the amount of the group 1 carbonate solution you add will only partially neutralise the sulfuric acid.

You will then complete the neutralisation of each mixture by titration with sodium hydroxide, **FA 2**.

In **1(c)**, you will use your volumes of **FA 2** and **FA 4** to plot a graph which will enable you to determine the concentration of the sulfuric acid solution and hence, determine the relative atomic mass of the element in the unknown group 1 carbonate, **FA 3**.

(a) Preparation of **FA 4**

1. Weigh accurately about 10.0g of **FA 3** into an empty weighing bottle. Record all weighings in the space provided. Make certain that your recorded results show the precision of your working.
2. Transfer the contents of the weighing bottle into a 250 cm^3 beaker and rinse the weighing bottle with deionised water and transfer the rinsing into the beaker. Repeat the rinsing to ensure all **FA 3** has been transferred.
3. Add 100 cm^3 of deionised water to dissolve **FA 3**.
4. Transfer **all** of the solution to a graduated flask and make up the solution to 250 cm^3 with deionised water and mix thoroughly.

Results

Mass of weighing bottle and FA 3 / g	
Mass of empty weighing bottle / g	
Mass of FA 3 / g	9.990

OR

mass of FA3 used (with Tare function) = 10.049 g

mass recorded to 3 dp + header + units + mass reading ranges from 9.950 to 10.049 g:

1 mark

[1]

(b) Preparation and titration of the reaction mixture

Notes: You will perform each titration **once** only. Great care must be taken that you do not overshoot the end-point.

You should aim not to exceed 25 minutes for this experiment.

1. Fill the burette labelled '**FA 2**' with **FA 2**.
2. Fill the burette labelled '**FA 4**' with **FA 4**.
3. Pipette 25.0 cm³ of **FA 1** into a 250 cm³ conical flask.
4. Add 6.00 cm³ of **FA 4** into the same conical flask from step 3.
5. Thoroughly swirl the mixture.
6. Add 2 to 3 drops of methyl orange indicator to the conical flask.
7. Titrate the mixture in the conical flask with **FA 2** until an orange colour is obtained.
8. Repeat steps 3 to 7 until four more aliquots using the following volumes of 9.50 cm³, 13.00 cm³, 17.00 cm³ and 20.00 cm³ of **FA 4** have been titrated. Record your results in the space provided. Make certain that your recorded results show the precision of your working.

Results

volume of FA 4 used / cm ³	6.00	9.50	13.00	17.00	20.00
final burette reading / cm ³	27.90	23.00	40.90	12.50	20.70
initial burette reading / cm ³	0.00	0.00	23.30	0.00	12.40
volume of FA 2 used / cm ³	27.90	23.00	17.60	12.50	8.30

[2]

- (c) Plot a graph of the volumes of **FA 2** added, on the y-axis, against the volumes of **FA 4** added on the x-axis on the grid in **Fig 1.1**.

You should choose scales for the axes which allow you to determine by extrapolation

- the volume of **FA 4** required, $V_{\max}(\text{FA 4})$, to completely react with 25.0 cm^3 of **FA 1** if no **FA 2** is added;
- the volume of **FA 2** required, $V_{\max}(\text{FA 2})$, to completely react with 25.0 cm^3 of **FA 1** if no **FA 4** is added.

Draw the line of best fit, taking into account all your plotted points. Hence obtain values for $V_{\max}(\text{FA 4})$ and $V_{\max}(\text{FA 2})$.

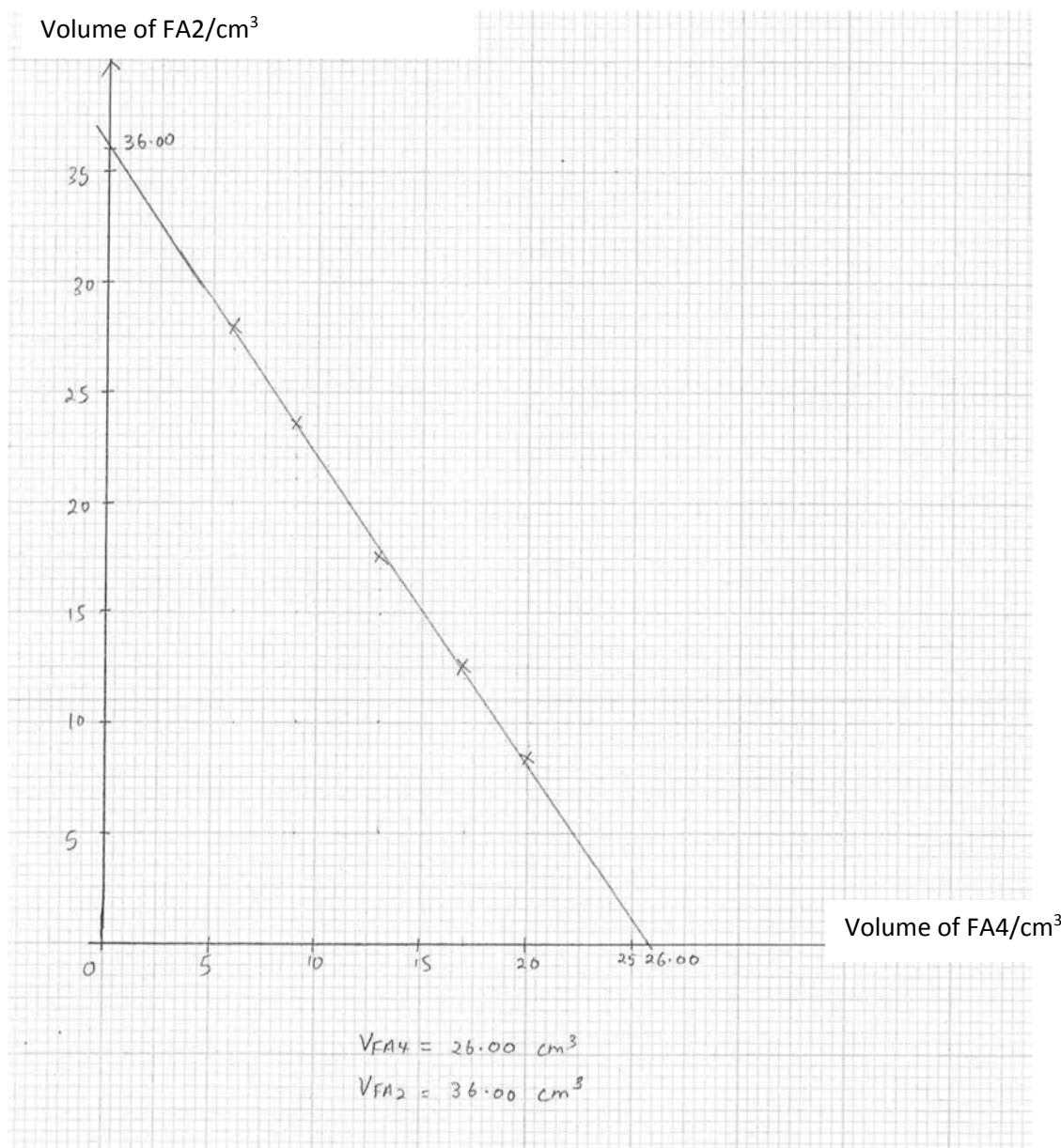


Fig 1.1

$V_{\max}(\text{FA 4}) = 26.00 \text{ cm}^3$

$V_{\max}(\text{FA 2}) = 36.00 \text{ cm}^3$

[7]

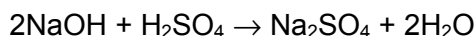
- (d) Suggest an explanation on the direction of the slope of your graph.

Negative gradient or downward sloping line as when volume of M_2CO_3 increases, volume of NaOH needed to neutralise remaining H_2SO_4 decreases.

[1]

Calculations

- (e) (i) Using appropriate data from your graph in (c), calculate the concentration of sulfuric acid in FA 1.



$$\frac{C_{H_2SO_4} \times V_{H_2SO_4}}{C_{NaOH} \times V_{NaOH}} = \frac{1}{2}$$

$$C_{H_2SO_4} = \frac{\frac{1}{2} \times 0.550 \times 36.00}{25.0} = 0.396 \text{ mol dm}^{-3}$$

Or

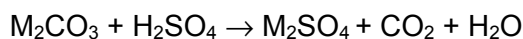
$$n_{NaOH} = 0.550 \times \frac{36.00}{1000} = 0.0198 \text{ mol}$$

$$n_{H_2SO_4} = \frac{1}{2} \times n_{NaOH} = \frac{1}{2} \times 0.0198 = 0.009900 \text{ mol}$$

$$C_{H_2SO_4} = \frac{0.009900}{\frac{25.0}{1000}} = 0.396 \text{ mol dm}^{-3}$$

[2]

- (ii) Using your answer in (e)(i) and appropriate data from your graph in (c), calculate the concentration of group 1 carbonate in FA 4.



$$\frac{C_{M_2CO_3} \times V_{M_2CO_3}}{C_{H_2SO_4} \times V_{H_2SO_4}} = \frac{1}{1}$$

$$C_{M_2CO_3} = \frac{0.396 \times 25.0}{26.00} = 0.38076 \text{ mol dm}^{-3} = 0.381 \text{ mol dm}^{-3}$$

OR

$$n_{H_2SO_4} = 0.009900 \text{ mol}$$

$$n_{M_2CO_3} = n_{H_2SO_4} = 0.009900 \text{ mol}$$

$$C_{M_2CO_3} = \frac{0.009900}{\frac{26.00}{1000}} = 0.38076 \text{ mol dm}^{-3} = 0.381 \text{ mol dm}^{-3}$$

[2]

- (iii) Hence, calculate the relative atomic mass of M.
[A_r: O = 16.0; C = 12.0]

$$n_{\text{M}_2\text{O}_3} = 0.38076 \times \frac{250}{1000} = 0.095190 \text{ mol}$$

$$M_r \text{ of } \text{M}_2\text{O}_3 = \frac{10.194}{0.095192} = 107.091 = 107.1$$

$$A_r \text{ of M} = \frac{107.091 - (12.0 + 16.0 \times 3)}{2} = 23.55 = 23.6$$

[2]

- (f) A student performs this experiment. Unknowingly, he uses a sample of **FA 3** which is slightly damp. State and explain the effect on the relative atomic mass of M calculated in e(iii).

Presence of moisture makes actual mass of M_2CO_3 added lower than expected.

Calculated concentration of M_2CO_3 or H_2SO_4 is higher and hence the A_r of M calculated is lower.

[2]

- (g) **Determination of concentration of sulfuric acid using gravimetric analysis**

A student performed an experiment to determine the concentration of sulfuric acid using its density. The student weighed the mass of an empty 250 cm³ volumetric flask and recorded the mass Table 1.1. He used a pipette to transfer 25.0 cm³ of sulfuric acid, **FA 1** into the flask and then re-weigh it. He then recorded the mass of the flask and **FA 1** in Table 1.1.

Table 1.1

Mass of flask and FA 1 / g	122.22
Mass of flask / g	96.62
Mass of FA 1 / g	25.60

- (i) Calculate the mass and hence the density of sulfuric acid in **FA 1** in g cm⁻³. Leave your calculated density to 3 decimal places.

$$\text{mass} = 25.60 \text{ g}$$

$$\text{density} = \frac{25.6}{25.0} = 1.024 \text{ g cm}^{-3}$$

[2]

- (ii) Use Fig. 1.2 and your answer in g(i), deduce the concentration of sulfuric acid in FA 1 in mol dm^{-3} .

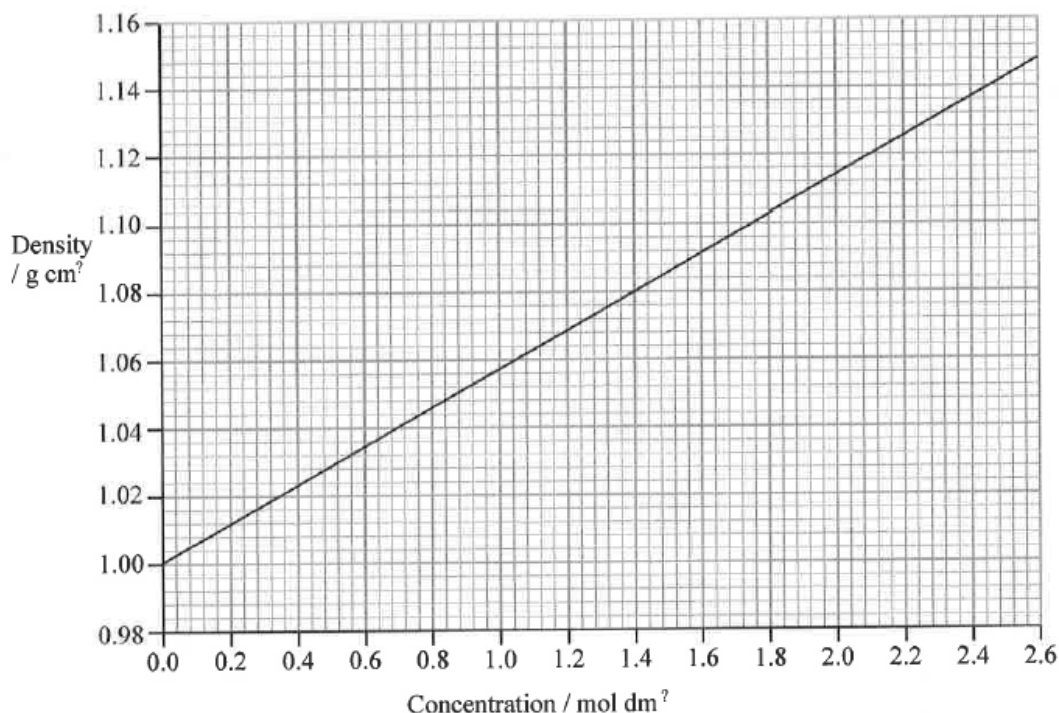


Fig. 1.2

concentration of sulfuric acid in FA 1 = $0.400 \text{ mol dm}^{-3}$
($0.38 - 0.42$, within $\frac{1}{2}$ square)

[1]

(h) **Determination of concentration of sulfuric acid using thermochemistry**

The neutralisation between a base and sulfuric acid is exothermic. Heat released is sufficient to allow the maximum temperature, ΔT_{max} , to be determined within a few minutes.

In this question, you are to perform an experiment to determine a value, ΔT_{max} . You will then use your value, ΔT_{max} to determine the concentration of sulfuric acid.

You will carry out the following instructions to obtain a value of ΔT_{max} .

1. Use a 25.0 cm^3 pipette, transfer FA 1 into a polystyrene cup held securely in a beaker. Record the temperature to 1 decimal place in the table provided.
2. Use a measuring cylinder, measure 100 cm^3 of FA 2. Record the temperature to 1 decimal place. Calculate the weighted average temperature, T_{ave} of T_{FA1} and T_{FA2} to 1 decimal place.
3. Add FA 2 carefully and quickly to FA 1 in the cup, stir and record the maximum temperature reached, again to 1 decimal place.

$$\text{Weighted average temperature, } T_{\text{ave}} = \frac{(V_{\text{FA1}} \times T_{\text{FA1}}) + (V_{\text{FA2}} \times T_{\text{FA2}})}{V_{\text{FA1}} + V_{\text{FA2}}}$$

Temperature of FA 1 before mixing, $T_{\text{FA1}} / ^\circ\text{C}$	31.0
Temperature of FA 2 before mixing, $T_{\text{FA2}} / ^\circ\text{C}$	29.0
Weighted average temperature, $T_{\text{ave}} / ^\circ\text{C}$	29.4
Maximum temperature reached, $T_{\text{max}} / ^\circ\text{C}$	32.0
$\Delta T_{\text{max}} / ^\circ\text{C}$	2.6

[2]

- (i) Use the equation below to find the concentration of the acid:

$$\text{Concentration} = 0.161 \times \Delta T_{\text{max}}$$

$$\text{Concentration} = \dots 0.419 \text{ mol dm}^{-3} \dots$$

[1]

- (ii) Instead of adding **FA 2** to **FA 1** to determine ΔT_{max} , the student added barium hydroxide, $\text{Ba}(\text{OH})_2$ of the same concentration and volume. Explain the impact on the value of ΔT_{max} .

Limiting reagent is H_2SO_4 although the base is changed from NaOH to $\text{Ba}(\text{OH})_2$.

The number of moles of water formed is the same.

Since the total volume after mixing **FA 1** and **FA 2** remains the same, the

amount of heat released is the same, ΔT_{max} is the same.

[2]

(i) Planning

The exothermic reaction between an acid and metal hydroxide can be used to determine the equivalence-point of a neutralisation reaction without the use of an indicator. This process is known as *thermometric titration* and can be used to calculate the concentration of an acid solution.

The procedure from **1(h)** could be further improved to obtain a more accurate value for the concentration of sulfuric acid in **FA 1**.

Plotting $\Delta T_{\max}$ against volumes of **FA 2** gives two best-fit lines. One line is drawn using data before the equivalence-point and the second line using the remaining data. These lines are then extrapolated until they intersect.

- (i)** Plan an investigation, based on the information above, to determine the equivalence-point for the neutralisation between **FA 1** and **FA 2** and hence the concentration of sulfuric acid in **FA 1**.

You may assume you are provided with

- 250 cm³ solution of dilute sulfuric acid, **FA 1**,
- 250 cm³ of 0.550 mol dm⁻³ sodium hydroxide, NaOH, **FA 2**,
- graph paper,
- the equipment normally found in the school or college laboratory.

In your plan you should include brief details of

- the apparatus you would use, bearing in mind the levels of precision they offer,
- the procedure that you would follow and the measurements that you would take,
- how you would recognise the equivalence-point had been passed,
- provide a sketch of the graph that you expect to obtain,
- how you would determine the concentration of sulfuric acid,
- state a safety precaution and suggest how do you minimise the risks involved.

1. Use a burette, transfer 30.00 cm³ of **FA 1** into a styrofoam cup placed in a beaker.

2. Use a thermometer, measure the initial temperature of **FA 1**.

3. Using a second burette, transfer 40.00 cm³ of **FA 2** into a second styrofoam cup placed in a beaker.

4. Use a thermometer, measure the initial temperature of **FA 2**.

5. Calculate the weighted average temperature of **FA 1** and **FA 2**.

6. Add **FA 2** into the Styrofoam cup containing **FA 1**. Cover the Styrofoam cup with the lid.

7. Use the thermometer to stir the mixture and measure the maximum temperature ($\Delta T_{\max}$) of the mixture using the thermometer

8. Note the value of $\Delta T_{\max}$

9. Wash the Styrofoam cup thoroughly and dry it

10. Repeat steps 1 to 8 using 10 cm³ of **FA 1** and 60 cm³ of **FA 2**, 20 cm³ of **FA 1** and 50 cm³ of **FA 2**, 40 cm³ of **FA 1** and 30 cm³ of **FA 2**, 50 cm³ of **FA 1** and 20 cm³ of **FA 2**.

11. Tabulate the volumes of **FA 1** and **FA 2**, and ΔT_{max}

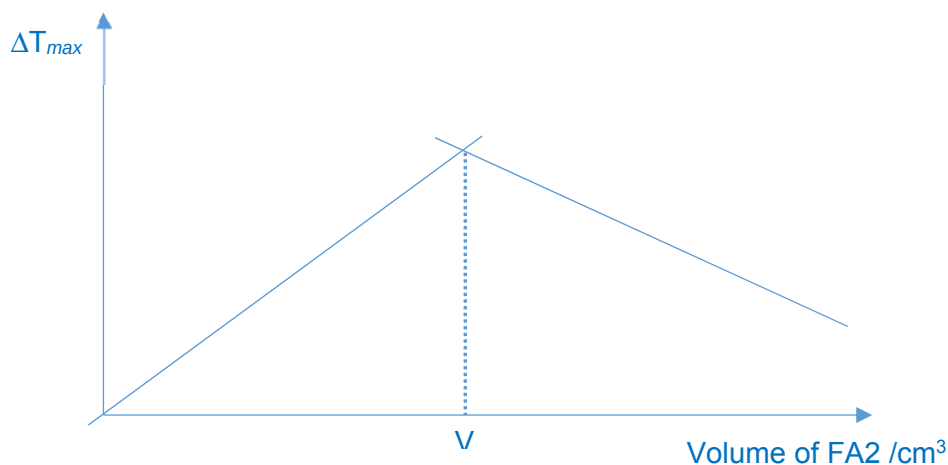
12. Plot the graph of ΔT_{max} against volume of FA2.

13. From the graph, the volume of **FA 2**, V which gives the highest ΔT_{max} is the exact volume of NaOH which is required to neutralise with H₂SO₄ from **FA 1**. The intersection of the two straight lines will be the neutralisation volume of H₂SO₄.

14. From the plot, when the ΔT_{max} begins to decrease, the neutralisation has passed.

$$15. C_{H_2SO_4} = \frac{1}{2} \times \frac{V_{FA2} \times 0.550}{V_{total} - V_{FA2} \text{ at equivalence}}$$

16. The H₂SO₄ or NaOH of high concentration is corrosive. Wear gloves to avoid skin contact.



[8]

- (ii) State and explain one consideration to be taken when deciding the minimum volume of **FA 1** to be used.

Enough solution to cover the bulb of thermometer so that the change in temperature of the solution could be measured quickly without heat lost.

OR

The volume of FA1 used must produce a significant change in temperature so that the percentage error of the thermometer can be kept low.

[1]

- (iii) Describe how you would use your graph in **1i(i)** to determine the enthalpy change of neutralisation between **FA 1** and **FA 2**.

Using maximum change in temperature, ΔT_{max} from the graph, calculate, heat of released from neutralisation using, $q = mc\Delta T$.

Using the volume of H_2SO_4 which causes ΔT_{max} , calculate no. of moles of

H_2SO_4 , n , and the number of moles of water is $2n$; Enthalpy of neutralisation, $\Delta H = - (q/2n)$.

[2]

- (iv) How, if at all, will ΔT_{max} vary if you use half the volumes of the original **FA 1** and **FA 2**. Explain your answer.

ΔH of neutralisation is constant. Half the volumes of original **FA 1** and **FA 2** will half the number of moles of water.

$$\Delta H_{\text{neutralisation}} = - \frac{\frac{1}{2}V \times C \times \Delta T}{\frac{1}{2}n \text{ of water}} ; \text{ with constant } \Delta H_{\text{neutralisation}}, \Delta T_{\text{max}} \text{ will be the same.}$$

[2]

- (v) Provide a reason why this method is preferred to that in **1(h)**.

This method makes use of more data points and is more accurate.

[1]

[Total: 41]

2 Investigation of some chemical reactions involving metal ions

FA 5 is a solid metal carbonate with the formula, XCO_3 .

FA 6 is an aqueous solution containing one cation and one anion.

- (a) Perform the tests described in Table 2.1, and record your observations in the table. Test and identify any gases evolved.

Table 2.1

	tests	observations
1	Place a spatula of solid FA 5 in a test-tube. Hold the test-tube in a holder. Heat the test-tube strongly until no further changes are seen. Keep the residue for test 2.	Green solid turned black upon heating. Gas evolved formed white ppt with limewater. Water condensed at upper part of test tube
2	To the residue from test 1, add sufficient amount of FA 1 to dissolve the residue completely. Keep this solution for tests 3 and 4.	Black solid dissolved to give a pale blue solution. (If solid from test 1 is not completely decomposed, effervescence will be seen here but not a marking point for this.)
3	Add ten drops of the solution obtained from test 2 to a test-tube. Add aqueous NH_3 to this test-tube, until no further change.	Blue ppt formed. Soluble in excess NH_3 to give a dark blue solution.
4	Add ten drops of the solution from test 2 to a test-tube. Then add about 1 cm depth of aqueous potassium iodide and shake the mixture thoroughly. To the mixture, add 1 cm depth of aqueous sodium thiosulfate.	White (off white) ppt formed in brown solution or brown ppt. White (off white) ppt in colourless solution (brown solution decolourised, white ppt seen).
5	Test the FA 6 solution with the universal indicator paper.	Universal indicator turned orange, pH = 3.
6	Add 1cm depth of FA 6 to a test tube. Then add a spatula of FA 3 to the test tube.	Effervescence observed and gas evolved formed white ppt with limewater. White ppt seen.
7	To 1 cm depth of FA 6 , add 10 drops of nitric acid followed by 10 drops of aqueous barium chloride.	White ppt formed.

[5]

(b) Consider your observation in **Table 2.1**.

- (i) Suggest the identities of the cations in **FA 5** and **FA 6** and the anion in **FA 6**.

Cation in **FA 5** : Cu^{2+}

Cation in **FA 6** : Al^{3+}

Anion in **FA 6** : SO_4^{2-}

- (ii) A student carried out test 1 using **FA 3**, it is unlikely for the same observation to be observed for this test. Suggest an explanation to account for this difference. [2]

FA 3 is a group 1 carbonate.

Charge density of group 1 cation is lower, hence it has lower polarising power.

Electron cloud of CO_3^{2-} is less polarised.

Thermal stability of group 1 carbonate is higher, hence does not get decomposed.

- (iii) Test 4 in **Table 2.1** shows a redox reaction to take place. Some relevant redox half-equations are given in Table 2.2. [2]

Table 2.2

half equation	E^θ / V
$\text{X}^{2+} + \text{e}^- \rightleftharpoons \text{X}^+$	+0.15
$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$	+0.54

Using the information in **Table 2.2**, calculate E^θ_{cell} for the reaction shown in your observation in test 4. Suggest a reason for the deviation from the prediction of E^θ_{cell} value calculated.

$$E^\theta_{\text{cell}} = +0.15 - (+0.54) = -0.39 \text{ V}$$

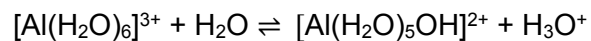
Reaction should not be spontaneous due to negative E^θ_{cell} .

Due the precipitation of CuI , $[\text{Cu}^+]$ dropped, equilibrium of $\text{Cu}^{2+} + \text{e}^- \rightleftharpoons \text{Cu}^+$ shifts right, favouring reduction.

$E^\theta_{(\text{Cu}^{2+}/\text{Cu}^+)}$ becomes more positive, E^θ_{cell} becomes more positive.

[2]

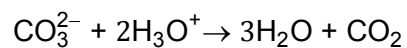
- (iv) Write an equation to account for the observation to test 5 in **Table 2.1**.



[1]

- (v) Use your answer to (iv) to account for the observations to test 6 in **Table 2.1**.

The formation of H_3O^+ from **FA 6** reacts with CO_3^{2-} from **FA 3** produces $\text{CO}_2(\text{g})$.



Further hydrolysis of Al^{3+} produces $\text{Al}(\text{OH})_3$ which accounts for the white ppt seen.

[2]

[Total: 14]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt. insoluble in excess	green ppt. insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt. insoluble in excess	off-white ppt. insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

<i>ions</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Test for gases

<i>ions</i>	<i>reaction</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	“pops” with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

END OF PAPER 4